

Evaluating Multimodal Alignment in ChatGPT and Gemini: An Empirical Study of Inter- and Intra-Model Consistency in Image Description and Generation

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Problem Statement and Motivation

- Intra-AI model: same AI model for both image description and generation
- Inter-AI model: different AI models for image description and generation
- Important for: trustworthiness, reproducibility, usability in creative, scientific and industrial contexts
- Research gap: lack of comprehensive analysis comparing intra-AI model and inter-AI model alignment in image description and generation tasks, combining human and automatic evaluation.

Research Questions

1. *How consistent, with respect to human preferences, are multimodal AI models in aligning descriptions and generations, both within the same model and across different models?*
2. *How well do automatic multimodal evaluation metrics align with human judgments?*

Related Work

- CLIP-AGIQA [Tang et al., 2024]
 - Automatic evaluation metric for image generation quality and alignment

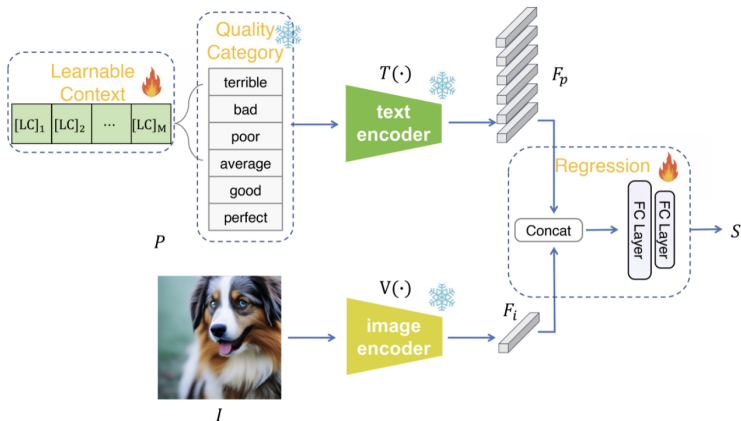
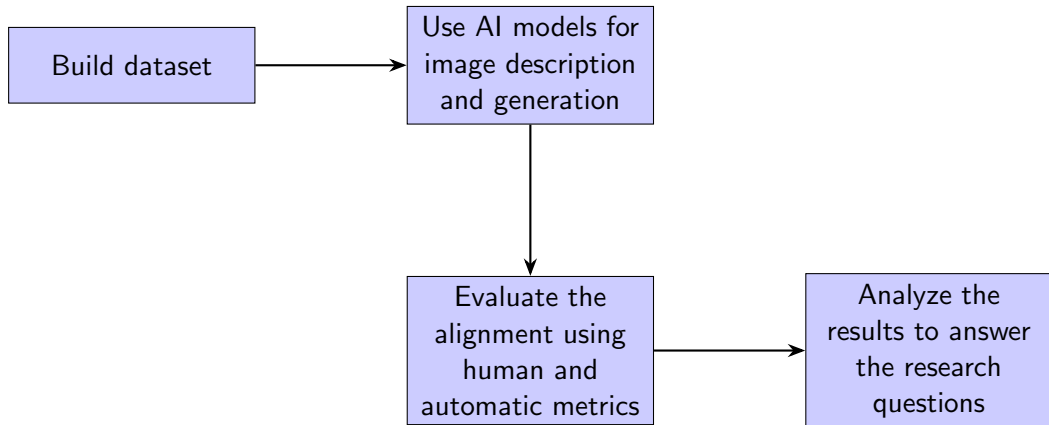


Fig. 2. Overall Architecture of *CLIP-AGIQA*.

Research Method



Dataset

- 10–15 images
- Collected explicitly for this study
- Objects, nature, people, text

Prompts

- Basic task description
- Detailed task description, specifying target audience
- Detailed task description, specifying target audience and their purpose

Evaluation Criteria

- Automatic metrics
 - CLIP-AGIQA
- Human judgments
 - 5–10 evaluators
 - Ranking output images based on provided reference

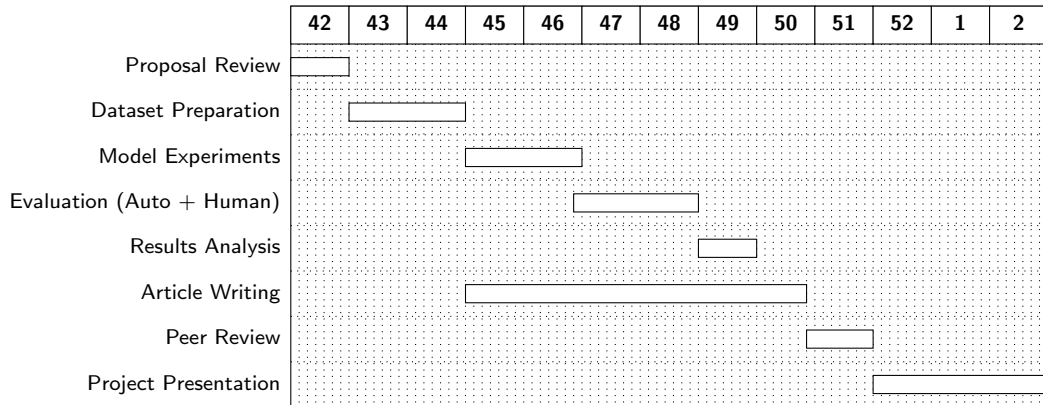
Results Analysis

- Inter-rater agreement
 - Kendall's W
- Correlation between human ranking and automatic metrics
 - Kendall's Tau
 - Spearman correlation

Expected Results

1. *Inter-AI model alignment will show significant divergence due to architectural heterogeneity, while intra-AI model descriptive vs. generative outputs will diverge less but still systematically*
2. *CLIP-based automatic metrics will partially but not fully capture human preferences, requiring complementary human evaluation*

Time Plan



References

 Tang, Z., Wang, Z., Peng, B., and Dong, J. (2024).

Clip-agiqa: Boosting the performance of ai-generated image quality assessment with clip.
In *International Conference on Pattern Recognition*, pages 48–61. Springer.

Thank You